BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

ZOOLOGY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Multiple Choice Questions)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. Malpighian tubules are characteristically associated with:

(a) Flat worm
(b) Insect
(c) Earth worm
(d) Round worm

Q2. It takes place in liver.

(a) Ornithine Cycle
(b) Bile Formation
(c) Cholesterol Formation
(d) All of these

Q3. Total number of bones in our four (4) limbs is:

(a) 30
(b) 60
(c) 90
(d) 120

Q4. Each myosin filament is surrounded by:

(a) 2 actin filaments
(b) 4 actin filaments
(c) 6 actin filaments
(d) 8 actin filaments

Q5. The hormone essential for physical and mental development of a child is:

(a) ADH
(b) Insulin
(c) Thyroxine
(d) FSH

Q6. The receptors for detection of sound energy are:

(a) Chemoreceptor
(b) Pain receptor
(c) Thermo receptor
(d) Mechano receptor

Q7. Ovulation is initiated by:

(a) FSH
(b) LH
(c) Estrogen
(d) Progesterone

Q8. Cytoplasmic localization is a consequence of:

(a) Fertilization
(b) Morula
(c) Cleavage
(d) Blastula

Q9. Which of the following is associated with fragile RBCs?

(a) Hemophilia
(b) Thalasemia
(c) Stickle cell Anemia
(d) All of these

Q10. Noise is considered as sound beyond.

(a) 40 decibels
(b) 80 decibels
(c) 20 decibels
(d) 30 decibels

Q11. Germinal continuity theory was proposed by:

(a) Hutton
(b) Lamarck
(c) Weismann
(d) Darwin

Q12. The enzymes that act like scissors in rDNA technolog are:

(a) Polymerase
(b) Ligase
(c) Reverse Transcriptase
(d) Restriction Enzyme

Q13. The part of brain related with long term memory is:

(a) Hippocampus
(b) Amygdala
(c) Limbic System
(d) Hypothalamus

Q14. It is considered as connecting link between nervous and endocrine system.

(a) Thalamus
(b) Hypothalamus
(c) Pituitary Gland
(d) Pineal Body

Q15. Which of the following is required for initiation of muscle fiber contraction?

(a) ATP
(b) Sodium
(c) Creatininc Phosphate
(d) Calcium

Q16. It some animal is either naturally or artificially prevented from breeding, it is called:

(a) Genetic Drift
(b) Gene Pool
(c) Genetic Death
(d) Gene Frequency

SECTION – B (SHORT ANSWER)

> Note: Answer any EIGHT of the following questions. Each question carries 02 marks.

Q2. Distinguish between any ONE of the following. (i) Asexual and Sexual Reproduction (ii) Lamarckism and Darwinism

Q3. Why excreting Ammonia is typically associated with freshwater inhibiting organisms?

Q4. Osteoporosis mainly affects female, wny?

Q5. Under strenuous exercise conditions, why skeletal muscles undergo fatigue?

Q6. What is "Emergency Hormone"? Why it is called so?

Q7. What are vestigial organs? Give four examples.

Q8. What do you understand by the term "endangered species"? Give any four examples from Pakistan.

Q9. Define any TWO of the following. (i) Protonephridia (ii) Restriction Enzyme (iii) Resting Membrane Potential (iv) Fossil Fuel

Q10. What is "Global Warming"? How it affecting living organisms?

Q11. How pyrexia helps in combating pathogenic microorganisms?

Q12. Differentiate between "Pivot Joint" and "Ball and Socket Joint".

Q13. Cleavage in chick is "Discoidal Type", why?

SECTION – C (DESCRIPTIVE ANSWER)

> Note: Answer any ONE of the following questions. Each question carries 10 (6 + 4) marks.

Q14. (a) Discuss the structure of skeletal muscle. (b) Explain how a hormone signals its target cell.

Q15. (a) State and explain the structure of the chick's egg and process of cleavage it goes through after fertilization. (b) How biotechnology is helpful in agriculture?